Explainable Predictive Analytics for Employee Attrition Risk Assessment

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***Abstract*—**Employee attrition creates replacement cost, disrupts project continuity, and removes organizational knowledge. Predictive systems can estimate risk, but opaque outputs offer limited support for responsible human-resource decisions. This paper presents an explainable employee-attrition decision-support system evaluated on the IBM HR Analytics benchmark of 1,470 records and 35 attributes. Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting are compared on a stratified 80/20 split. Random Forest achieves 82.65% accuracy, 45.24% precision, 40.43% recall, a 42.70% F1-score, and 78.55% ROC-AUC on 294 held-out records. It is deployed as an implementation choice with direct TreeSHAP compatibility. Global SHAP analysis identifies overtime, stock-option level, age, monthly income, and manager tenure as the strongest modeled contributors. A React and FastAPI application integrates aggregate monitoring, local explanations, counterfactual simulation, and retention-action logging. The system supports human review and does not infer causation or automate employment decisions.

Keywords—employee attrition, explainable artificial intelligence, SHAP, random forest, HR analytics, decision support.

I. INTRODUCTION

Voluntary employee departure affects cost, delivery capacity, and institutional memory. Human-resource analytics therefore uses supervised learning to identify records that resemble prior attrition cases. A probability alone, however, does not explain why a model assigned risk or what response would be appropriate. Explainability connects a prediction to feature contributions [2]–[4], while human oversight remains necessary because statistical association cannot determine an employee's intention or justify adverse action.

This work treats attrition prediction as a decision-support problem. It combines controlled model comparison, global and local SHAP explanations, and a full-stack application for risk review, counterfactual testing, and retention-action tracking. The empirical claims remain restricted to the supplied benchmark and stored model artifacts.

A. Contributions

The study contributes: 1) a reproducible comparison of four classifiers on the same held-out set; 2) a deployment decision that considers minority-class behavior and explanation compatibility; 3) TreeSHAP explanations embedded in an operational review workflow; 4) a simulator that re-scores job-related changes without modifying the source record; and 5) explicit responsible-use limits.

B. Research Questions

RQ1 examines model differences across accuracy, precision, recall, F1-score, and ROC-AUC. RQ2 identifies the highest global SHAP contributions. RQ3 examines how model evidence can support a human-in-the-loop HR workflow without overstating benchmark findings.

C. Problem Definition and Scope

The prediction task is binary classification of a benchmark label. Let x denote an employee feature vector and y denote the observed Attrition value, encoded as one for Yes and zero for No. The estimator returns a score s(x) associated with the positive class. A threshold converts this score into a predicted label, while a separate set of thresholds assigns interface categories. Neither representation includes a time-to-event estimate. Consequently, the system cannot determine whether an employee will resign within a month, a quarter, or any other specified horizon. This distinction is relevant when interpreting the term risk assessment in the project title.

There are three different outputs in the application: a statistical estimate, a feature explanation, and a suggested review activity. Each has a different evidential basis. The estimate comes from a fitted classifier. The explanation decomposes that classifier's output. The recommendation comes from explicit rules in the application. A salary-review recommendation is therefore not an experimentally proven treatment effect. Keeping these outputs distinguishable allows the application to communicate useful information without attributing additional authority to a model-generated display.

D. Evaluation Perspective

The paper examines a supplied implementation and its persisted experiment results. It reports stored model performance and derives additional descriptive measures from the recorded confusion matrices. The analysis does not constitute a fresh training run or a prospective deployment trial. This scope supports a system-oriented research contribution: connecting model evidence to an inspectable software workflow and identifying where further validation is necessary. The original project presentation supplies the application context, while the

training script and stored metrics supply the methodological and quantitative details.

The practical user is an HR reviewer who needs to understand how an estimate was produced and what uncertainty accompanies it. Such a user may review several employees because a model has identified a pattern, but must still establish whether the underlying records are accurate and relevant. This motivates presenting the prediction together with its explanations, model version, assessment date, and review history. A transparent workflow is useful even when the classifier has moderate minority-class performance, provided its limits remain visible.

II. RELATED WORK

Random Forest aggregates randomized tree predictors to reduce variance [1], while gradient boosting constructs an additive model through stage-wise optimization [5]. Such ensembles can model nonlinear interactions in tabular HR data, but their structure complicates interpretation. SHAP provides a unified additive attribution framework [2], and TreeSHAP computes exact, consistent contributions for tree ensembles [3]. LIME offers a model-agnostic local surrogate approach [4]. Covert et al. place these methods within a broader family of removal-based explanations [14].

Attrition studies commonly compare linear, tree, and ensemble classifiers and identify overtime, compensation, tenure, and satisfaction as important predictors [8], [11]–[13]. Explainable HR decision-support research combines SHAP with what-if analysis [9]. Knowledge-graph approaches show that relational features may improve turnover prediction while retaining explainability [10]. Recent work also demonstrates that class-imbalance treatment can substantially change minority detection [15].

A. Predictive Models and Explanation Methods

The literature motivates a distinction between prediction quality and explanation quality. A model can rank observations well while producing a poor fixed-threshold classifier. It can also produce a mathematically valid explanation that remains difficult for an HR practitioner to interpret. In this implementation, predictive comparisons use the same test partition, and the explanation mechanism is selected to match the deployed model family. This makes the experiment internally comparable while leaving user understanding and explanation usefulness as separate questions requiring empirical assessment.

The foundational model papers [1], [5] explain the learning procedures rather than establishing an advantage for this particular dataset. Similarly, the SHAP and LIME papers [2]–[4] establish methods of interpreting a fitted function, not the causal validity of employment recommendations. Their role here is methodological. The present experiment must still demonstrate its own error profile, and the interface must describe explanations in language that reflects the model output actually being explained.

B. Relationship to HR Decision Support

Employee-attrition research provides the direct application context. The contribution of HR-DSS [9] is especially relevant because it combines explanations and what-if exploration. The present prototype uses the same broad decision-support motivation but implements a specific React and FastAPI workflow with stored predictions and retention records. The domain studies [8], [10]–[13] motivate comparison of several classifiers and investigation of operational variables. Their reported scores are not included in the performance graph, because differences in splits, preprocessing, resampling, and task definitions prevent a fair direct ranking against the current implementation.

Imbalanced-learning research [6], [15] motivates attention to the rare class. SMOTE is one possible resampling strategy, but it is not used in the supplied training pipeline. Instead, three of the four classifiers use balanced class weights. The distinction matters because weighting and generating synthetic observations are different interventions on the training problem. A reproducible report should state which procedure was executed rather than imply that all techniques discussed in related work belong to the implementation.

C. Position of the Present Work

The research gap addressed here is integration and traceability. A classifier comparison alone does not show how an analyst will inspect an employee, test an adjustment, and record a follow-up action. Conversely, a visually polished dashboard does not establish predictive validity. The paper links the analytical pipeline to the application data model and documents the limits of each connection. It makes no claim that the Random Forest algorithm, SHAP, or counterfactual input editing is a new invention. Its contribution is a concrete implementation and a critical examination of the evidence supporting that implementation.

III. METHODOLOGY

A. Dataset and Preprocessing

The IBM HR Analytics Employee Attrition and Performance benchmark contains 1,470 records and 35 attributes. Predictors cover role, department, compensation, work patterns, satisfaction, tenure, and related operational variables. Categorical attributes undergo one-hot encoding, and numeric features pass through the stored preprocessing pipeline, including StandardScaler where configured. The experiment uses an 80/20 stratified partition with seed 42, producing 294 held-out records.

B. Candidate Models and Metrics

The comparison includes Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting. Accuracy measures total correctness, precision measures the reliability of predicted attrition, recall measures detection of actual attrition, F1-score balances precision and recall, and ROC-AUC evaluates ranking across thresholds. Multiple metrics are necessary because attrition is the minority class.

TABLE I. MODEL CONFIGURATION

Model	Configuration
Logistic Reg.	lbfgs; balanced; max_iter=1000
Decision Tree	gini; depth=5; balanced
Random Forest	160 trees; depth=8; balanced
Gradient Boost.	130 estimators; lr=0.08; depth=4

C. Feature Construction and Leakage Control

The source pipeline drops EmployeeCount, Over18, StandardHours, and EmployeeNumber, together with the Attrition target, before fitting. This leaves 30 raw input fields. EmployeeNumber is retained for display and record identity but does not enter the model. The categorical list comprises BusinessTravel, Department, EducationField, Gender, JobRole, MaritalStatus, and OverTime. All remaining predictors are treated as numerical. Integer-coded satisfaction and education variables therefore enter the numerical branch rather than being one-hot encoded.

The ColumnTransformer fits only on the training partition. Its numerical branch uses StandardScaler, and its categorical branch uses OneHotEncoder with drop=first, sparse_output=False, and handle_unknown=ignore. The fitted transformer subsequently processes the test partition and the full dataset. Fitting the encoder and scaler after the split prevents information from the test-set distribution entering the learned transformation. Unknown categorical levels are accepted, but their resulting encoding must be interpreted carefully because an all-zero pattern can resemble the reference category.

D. Training and Selection Procedure

All four models receive the same transformed training records and use random_state=42 where available. Logistic Regression, Decision Tree, and Random Forest set class_weight=balanced. Gradient Boosting uses its specified estimator count, learning rate, and depth without that class-weight argument. These settings form the experiment configuration; the source does not document a hyperparameter search or nested cross-validation. Claims of optimal parameter selection would therefore exceed the evidence.

The training script explicitly assigns best_name to Random Forest before fitting the candidates. It then saves that estimator after computing all model metrics. The deployed model is consequently a preselected implementation choice, not an automated winner of a validation competition. The code's stated motivation is balanced behavior and TreeSHAP support. However, Gradient Boosting is also a tree ensemble, so TreeSHAP compatibility alone does not prove Random Forest is uniquely suitable. Generalization stability would require repeated runs or additional validation, which is outside the stored experiment.

E. Metric Definitions and Units

For true positives TP, true negatives TN, false positives FP, and false negatives FN, accuracy equals (TP+TN)/N. Precision equals TP/(TP+FP), recall equals TP/(TP+FN), and F1 equals 2TP/(2TP+FP+FN). Specificity equals TN/(TN+FP). Balanced accuracy averages recall and specificity, giving equal weight to the two classes even when their frequencies differ. All percentage-valued metrics in the tables use these definitions with Attrition=Yes as the positive class.

ROC-AUC uses continuous positive-class scores. It cannot be reconstructed from a single confusion matrix, because that matrix describes only one threshold. Accordingly, the paper reports the persisted AUC values but does not fabricate ROC curves. The precision-recall scatter plot likewise contains four operating points rather than complete precision-recall curves. These plotting choices preserve the distinction between quantities actually available in the artifact and quantities requiring per-observation scores.

IV. SYSTEM ARCHITECTURE

The system uses five connected layers. The data layer stores the benchmark and derived records. The preprocessing layer produces model-ready features. The modeling layer compares classifiers and persists the selected estimator. The explainability layer calculates global rankings and employee-level SHAP contributions. The application layer exposes evidence through FastAPI endpoints and a React interface.

TABLE II. SYSTEM LAYERS

Layer	Technology	Output
Data	CSV; SQLite	Employee and action records
Preprocessing	One-hot; scaling	Model matrix
Modeling	scikit-learn	Risk probability
Explainability	TreeSHAP	Global/local attribution
Application	FastAPI; React	Dashboard and simulator

Dedicated API routers provide aggregate statistics, employee lookup, model metadata, simulation, and retention actions. The simulator copies an employee feature vector, applies explicit overrides, transforms the simulated row, and returns the revised probability, tier, risk difference, and leading SHAP contributions.

A. Training Path and Serving Path

The architecture contains an offline analytical path and an interactive serving path. Offline processing reads the CSV, fits transformations, trains the candidates, saves the selected estimator, computes SHAP values, and populates the database. The serving path reads employee records and stored predictions or executes a new simulation with the saved artifacts. This separation avoids retraining the model for every dashboard request and keeps the interface responsive in principle, although no latency benchmark is provided.

The full dataset is transformed after the training-only fit and is used to create demonstration predictions for every record. These application-wide predictions include both training and held-out observations. Dashboard counts based on that population must therefore not be treated as an independent performance evaluation. Only the 294-record holdout supports the classifier comparison. This distinction also applies to the global SHAP ranking, which aggregates explanations across the full set of 1,470 records rather than only unseen observations.

B. Persistence and Record Identity

Employee records contain the benchmark attributes and display metadata. Prediction records link a probability, predicted label, risk category, baseline SHAP value, leading signed factors, recommendations, and assessment timestamp to an employee. Retention records store action type, title, priority, status, assignment, notes, and target date. ModelEvaluation records preserve the comparative metrics and hyperparameter metadata. Together these entities connect an interface record to an analytical output, but they do not yet provide immutable model-version lineage.

The seeding routine generates illustrative names, email addresses, avatars, and example retention actions. These values support the software demonstration and should not be interpreted as a study of identifiable employees. In particular, a seeded action marked Completed does not show that an intervention was delivered or evaluated. The benchmark and synthetic display identities must remain conceptually separate when presenting application evidence.

C. Service Interfaces

The employee listing supports pagination, department, role, risk, and overtime filters, text search, and restricted sorting options. The detail endpoint gathers employee attributes, prediction evidence, and retention actions. The retention API provides listing, creation, and update operations, with a record-existence check before creating an action. The health endpoint returns static service metadata and dataset information. It is a service-availability indicator and does not test model freshness, database integrity, or explanation correctness.

The FastAPI application registers separate routers and permissive cross-origin settings for development. The supplied routes do not demonstrate authentication or per-user authorization. Consequently, authorized-reviewer behavior is a deployment requirement rather than a property proven by this prototype. The software is best characterized as a local academic implementation whose analytical functions can be inspected before operational hardening.

V. EXPERIMENTAL RESULTS

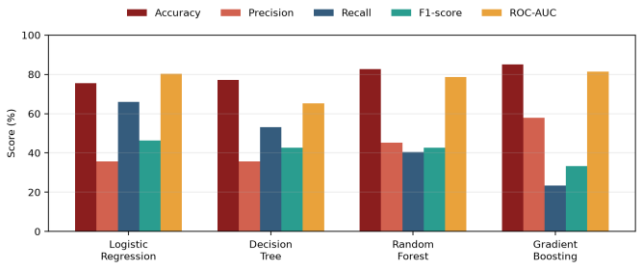


Fig. 1. Held-out performance for four candidate models (N=294).

TABLE III. HELD-OUT MODEL PERFORMANCE (%)

Model	Acc.	Prec.	Rec.	F1	AUC
Log. Reg.	75.51	35.63	65.96	46.27	80.35
Decision Tree	77.21	35.71	53.19	42.74	65.17
Random Forest	82.65	45.24	40.43	42.70	78.55
Grad. Boost.	85.03	57.89	23.40	33.33	81.46

Gradient Boosting records the highest accuracy and ROC-AUC, but its 23.40% recall means it misses most actual attrition cases at the stored threshold. Logistic Regression achieves the highest recall and F1-score while generating more false positives. Random Forest occupies a middle position and integrates directly with TreeSHAP. The deployed choice is therefore a system-design tradeoff rather than a claim that Random Forest leads every metric.

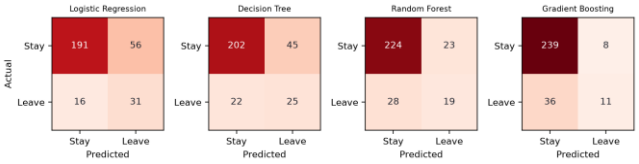


Fig. 2. Confusion matrices on the same held-out records.

Logistic Regression detects 31 of 47 actual attrition cases but produces 56 false positives. Gradient Boosting produces only eight false positives but detects 11 actual cases. Random Forest detects 19 cases with 23 false positives. These results show why accuracy alone is insufficient for deployment decisions.

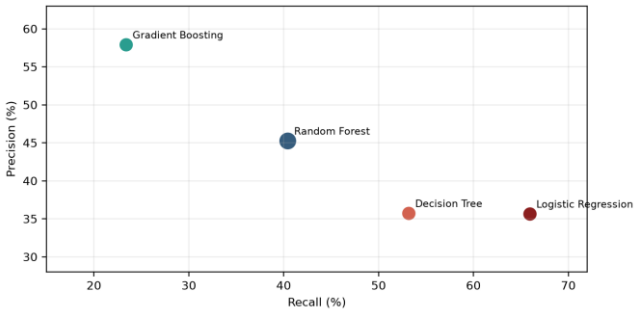


Fig. 3. Precision–recall positions at the stored decision threshold.

A. Majority-Class Baseline

The recorded matrices each contain 247 negative and 47 positive test examples, for a positive prevalence of 15.99%. A classifier predicting Stay for every example would achieve 247/294 = 84.01% accuracy and zero recall for attrition. Random Forest's 82.65% accuracy is below this baseline, while Gradient Boosting's 85.03% is only about 1.02 percentage points above it. These facts do not make the models useless: the baseline detects no leavers. They demonstrate why a claim of high accuracy alone is inadequate for this task.

Balanced accuracy calculated from the stored matrices is approximately 71.64% for Logistic Regression, 67.49% for Decision Tree, 65.56% for Random Forest, and 60.08% for Gradient Boosting. Unlike ordinary accuracy, this measure does not reward the classifier simply for the majority-class prevalence. The ordering is consistent with the stronger minority sensitivity of the weighted linear and single-tree baselines. The derived measures supplement the recorded metrics without claiming a new experiment.

B. Review Workload and Missed Cases

The models flag 87, 70, 42, and 19 test records, respectively, for Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting. Random Forest's 42 flags contain 19 true positives and 23 false positives. Thus, more than half of its positive predictions correspond to recorded stayers at this operating point. A

reviewer must understand that prioritizing those records means accepting a substantial number of unnecessary reviews. The 28 false negatives simultaneously show that the model misses a majority of the actual positive cases.

Compared with Gradient Boosting, Random Forest detects eight additional positive examples but introduces 15 additional false positives. Compared with Logistic Regression, it avoids 33 false positives but misses 12 positives that the linear model detects in aggregate. These are count comparisons, not claims that the same individuals are shared across error sets; establishing overlap would require per-observation predictions. The counts nevertheless make the operational tradeoff concrete.

C. Error-Cost Sensitivity

Let c_FP be the cost of an unnecessary review and c_FN the cost associated with a missed positive case. A simple retrospective error cost is c_FP times FP plus c_FN times FN. The recorded costs are 56c_FP+16c_FN for Logistic Regression, 45c_FP+22c_FN for Decision Tree, 23c_FP+28c_FN for Random Forest, and 8c_FP+36c_FN for Gradient Boosting. Random Forest has a lower cost than Gradient Boosting only when c_FN/c_FP exceeds 15/8, or 1.875. Logistic Regression becomes less costly than Random Forest when the ratio exceeds 33/12, or 2.75.

These break-even ratios are arithmetic consequences of the matrices. They do not establish monetary values for attrition or prescribe an employer's priorities. They also omit costs of inappropriate intervention, employee distrust, and constraints on reviewer capacity. Their purpose is to show that no single model is best without an explicit decision objective. A future study should define that objective before selecting a classifier or tuning a threshold.

D. Sampling Uncertainty

There are only 47 positive examples in the held-out sample. Detecting one additional positive would change recall by about 2.13 percentage points. Small absolute count differences can therefore create apparently meaningful percentage differences. Repeated stratified splits, paired prediction comparisons, and confidence intervals would be needed to establish whether observed model differences persist. The present results should be read as a descriptive snapshot of one partition, not a statistically demonstrated superiority claim.

VI. EXPLAINABILITY ANALYSIS

SHAP represents each prediction as a baseline plus additive feature contributions [2]. Global importance aggregates the magnitude of contributions across records, while local explanations retain the direction and magnitude for one employee. TreeSHAP enables efficient computation for the deployed ensemble [3].

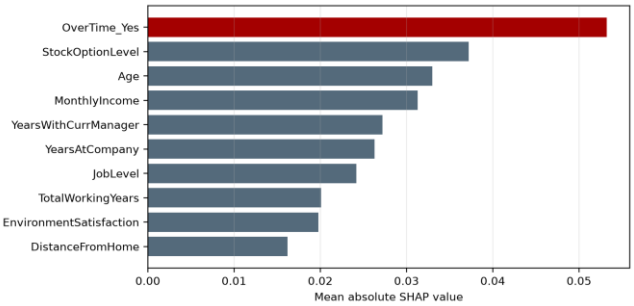


Fig. 4. Ten highest global mean absolute SHAP values.

Overtime is the strongest global feature, followed by stock-option level, age, monthly income, and years with the current manager. The ranking describes model behavior in this dataset; it does not prove that changing a feature will cause retention. Correlation, dataset construction, and the fitted model can all affect attribution.

A. Signed Contributions and Global Magnitude

For the explained output, the additive relation is f(x)=phi_0+sum_j phi_j(x), where phi_0 is the explainer baseline and phi_j is the contribution attributed to feature j. Global importance is the average absolute contribution over the chosen population. Taking the absolute value removes sign. Consequently, the stored overtime importance of 0.0532 cannot by itself be described as a positive 5.32-percentage-point effect for every employee. A local signed value is required for a directional statement.

The implementation sorts contributions by magnitude and stores up to five positive and five negative factors per employee. This is useful for concise displays but does not show the entire decomposition. The displayed top factors may not sum to the prediction-minus-baseline difference because omitted features retain contributions. A complete explanation view should either include the remaining contribution or identify the list as a selected subset. Otherwise, a user may incorrectly conclude that the visible bars account for the full score.

B. Feature Representation

The categorical encoding produces named indicator features, whereas numerical fields retain their original names after scaling. A displayed label such as OverTime_Yes refers to an encoded model input. Interpretation of dropped reference categories requires the preprocessing metadata. For readable HR explanations, the interface should map transformed names back to original fields and human-readable values while preserving the numerical attribution and its sign.

The saved SHAP baseline is a model-explanation reference, not necessarily the empirical turnover prevalence. Class weighting changes fitted tree behavior, and the explainer's reference construction also matters. A baseline near one-half should not be described as evidence that half the workforce is expected to resign. The distinction between model output, observed outcome frequency, and explanation baseline is essential to avoiding misleading risk communication.

C. Correlation and Explanation Stability

Several benchmark fields encode related concepts, including age, total experience, company tenure, and years with a manager. Attribution among correlated inputs depends on the fitted decision structure and the explanation assumptions. A high rank for manager tenure cannot isolate the causal effect of a manager relationship. Likewise, salary, job level, and role may share predictive information. The ranking remains informative about this model but does not identify an independent intervention target.

No explanation-stability experiment is reported. Future evaluation should examine whether small plausible input changes or training perturbations substantially alter the leading explanations. The metric of interest would include both score variation and changes in the order or sign of important factors. This evaluation would complement, rather than replace, human assessment of whether the explanation is understandable and appropriate for the intended review task.

VII. DECISION-SUPPORT APPLICATION

A. Executive Risk Radar and Directory

The dashboard presents evaluated records, historical attrition, average predicted risk, risk-tier distribution, department summaries, overtime comparisons, and vulnerable-role summaries. The directory supports filtering by role, overtime status, and Low (<30%), Medium (30–59%), or High (≥60%) modeled risk. Selecting a record opens the probability and local SHAP explanation together.

B. What-If Simulator

The simulator can vary overtime, monthly income, work-life balance, satisfaction, stock-option level, distance from home, job involvement, salary-hike percentage, and years in the current role. It returns original and simulated scores, category changes, applied adjustments, and leading simulated SHAP values. A lower simulated score describes the fitted model under modified inputs; it is not a causal estimate of an intervention.

C. Retention Action Tracking

The interface allows reviewers to record supportive actions and follow-up dates. Examples include workload review, compensation parity analysis, and manager check-ins. Logging documents a human decision but does not validate that an action caused later retention.

D. Category Semantics and Human Review

The interface thresholds differ from the classifier's decision threshold. A probability between 0.50 and 0.60 is a positive classifier prediction but remains in the Medium display tier. The High tier therefore cannot be substituted for the positive class when interpreting Table III. Category boundaries are application conventions rather than empirically validated levels of urgency. The word calibrated should be avoided because the supplied experiment does not provide a calibration curve, Brier score, or calibration procedure.

An appropriate review begins by verifying the record and checking its assessment time. The reviewer then examines the signed factors and the employee's context, rather than assuming a model score establishes dissatisfaction. The next step is an optional supportive conversation. Any proposed activity must be assessed for feasibility and consent before it is recorded. The action center can document this sequence, but the presence of an action record alone cannot demonstrate that these steps occurred.

E. Simulator Semantics and Edge Cases

For an employee selected by identifier, the simulation begins with the database record and compares the modified prediction with the stored baseline. If the stored prediction was generated by a different model version, the difference can mix a model update with an input change. A consistent implementation should re-score both the original and modified vectors with the same loaded artifacts. The current artifact set does not record sufficient immutable provenance to exclude that mismatch in every future use.

When no employee identifier is supplied, the source uses a default profile and a fixed original score of 0.55. This default comparison should be described as illustrative rather than a measured baseline effect. A reliable counterfactual calculation would first score the unmodified default profile. The API's risk-reduction field multiplies the probability difference by 100, so its unit is percentage points, even though the field name uses the word percentage. A reduction from 0.60 to 0.45 is 15 percentage points, whereas the relative percentage reduction would be 25%.

The simulator changes variables independently and accepts generic custom overrides. Some combinations can be unrealistic: altered tenure may contradict total experience, and changed income may conflict with role and job level. Before deployment, allowed inputs should enforce plausible ranges, cross-field constraints, and a restricted list of editable attributes. The presentation's hybrid-work example is not a dedicated measured hybrid-work feature in the training data. It should be understood as a proposed supportive activity, not a separately validated causal scenario.

F. Recommendation Rules

The source recommendation routine examines selected feature names and current attribute values. It produces up to three predefined activities, such as compensation review or workload rebalancing. These are rule-based suggestions rather than a learned treatment recommendation system. In addition, the feature-name check for OverTime does not match the transformed OverTime_Yes label, although the routine also checks the raw overtime value. This illustrates why feature naming and rule mapping need tests before the explanation-to-action link can be trusted.

VIII. IMPLEMENTATION AND REPRODUCIBILITY

The backend uses Python FastAPI, SQLAlchemy, and SQLite. Serialized artifacts include the selected Random

Forest, preprocessing pipeline, SHAP explainer, and metrics file. The frontend uses React, Vite, and Tailwind CSS. REST endpoints separate statistics, employees, model evidence, simulation, and retention actions.

TABLE IV. REPRODUCIBILITY ARTIFACTS

Artifact	Purpose
best_model.joblib	Selected Random Forest
preprocessor.joblib	Feature transformation
shap_explainer.joblib	TreeSHAP computation
metrics.json	Measured results
attrition_system.db	Application records

The project retains the holdout size, random seed, model parameters, metrics, and confusion matrices. These artifacts support regeneration of the reported figures. Reproducibility does not establish external validity; another organization, time period, or class distribution may yield different behavior.

A. Artifact Compatibility and Reproducibility

The preprocessor, model, and explainer form a coupled artifact set. The model expects exactly the transformed column order used during training, and the explanation labels must match that same order. Preserving only the estimator is insufficient. Package versions, training configuration, transformed feature names, and an artifact identifier should accompany each release. The current project retains basic configuration and metrics, but a complete reproducibility record would add dependency locks and hashes for the dataset and all serialized artifacts.

Inference should obtain transformed names from the fitted ColumnTransformer rather than infer them from a newly constructed row. The simulation code reconstructs numerical and categorical names from the incoming dataframe, whereas the transformer holds its training-time column order. Any disagreement can mislabel SHAP values even when the prediction itself is computed correctly. A parity check should compare the persisted transformed names, transformer output dimensions, model input dimensions, and explainer contribution dimensions.

B. Functional Verification Protocol

A reproducible verification sequence should first confirm that every matrix sums to the reported holdout size and that its derived precision, recall, F1, and accuracy agree with the stored values. It should next verify that an unchanged employee vector receives the same score in offline and online paths. A zero-adjustment simulation should yield zero probability difference when the same artifact set is used. These tests check consistency; they do not measure out-of-sample generalization.

For explanations, the full contribution vector should reconstruct the explained output within numerical tolerance. Feature labels should align with transformed columns, and filtering the visible factors should preserve their signs. API tests should verify missing-employee responses, pagination boundaries, category filters, and persistence of action updates. A database health check should detect missing relationships between employees and predictions instead of returning a default low-risk interpretation.

C. Operational Boundaries

The training entry point drops and recreates database tables before seeding the demonstration. Re-running it would erase existing action records, so training and operational persistence must be separated before live use. A migration process and independent experiment database would avoid coupling retraining with retention-history destruction. The paper records this as an implementation constraint rather than implying that the provided seeding script is a safe deployment procedure.

No throughput, tail latency, uptime, browser usability, or multi-user concurrency benchmark is available. The paper therefore describes architecture and source behavior without assigning production-readiness status. Future performance testing should measure model loading, preprocessing, prediction, and explanation separately, then report end-to-end latency under defined concurrent load. Such measurements would identify whether caching and stored explanations provide the expected operational benefit.

IX. RESPONSIBLE AI AND LIMITATIONS

Employment decisions can materially affect individuals. The system is limited to supportive decision assistance and must not automate firing, promotion denial, or discipline. Employee-level risk data requires role-based access, audit logging, encryption, retention limits, and a correction process. Explanations improve visibility into model behavior but do not make an unfair model fair or a correlation causal.

The study has several limitations. It uses a public benchmark rather than a live organizational cohort; reports a single stratified holdout rather than repeated or temporal validation; does not test subgroup fairness, probability calibration, or deployment drift; and uses SHAP attributions that depend on the fitted model and background assumptions. The simulator changes inputs independently and does not estimate causal effects.

A. Sensitive Inputs and Fairness Evidence

The training script includes Gender, MaritalStatus, and Age. Thus, the implementation does not support a claim that all sensitive demographic attributes were excluded. These fields and their proxies require a task-specific fairness assessment before any employment use. Merely hiding them in the interface would not remove their influence on the classifier, and deleting them would not necessarily eliminate indirect effects through correlated variables. This paper reports their presence explicitly so the governance discussion matches the implemented pipeline.

An appropriate fairness study would compare positive prediction rates, false-positive rates, false-negative rates, and calibration across adequately sized groups. Those measures can conflict, particularly when observed base rates differ. The choice of criterion should follow the intended use and institutional review, not be selected after inspecting which metric looks most favorable. The existing artifacts do not contain this analysis, and no

fairness conclusion can be drawn from the overall accuracy or the global SHAP ranking.

B. Data Quality and Temporal Validity

Benchmark records are static. They cannot establish whether a variable precedes resignation, is updated after a departure decision, or would be available at the intended prediction time in a real HR system. Before operational training, each field should be assigned an observation timestamp and eligibility rule. Information recorded after the outcome must be excluded. Temporal validation should then test whether performance persists on later cohorts, because a random split alone cannot assess organizational or economic drift.

The target also does not distinguish resignation motivations, involuntary separation, or preventable turnover within a defined time window. A general attrition label may combine outcomes that require different responses. A future data collection protocol should specify which departures are relevant, how censoring is handled, and how long employees are followed. Without those definitions, a benchmark model cannot be interpreted as a comprehensive retention-planning instrument.

C. Interface and Organizational Risks

Risk labels can influence how managers perceive employees even when the underlying score is uncertain. Restricting access and avoiding punitive action is therefore part of the system's intended governance, not an optional decoration. Reviewers should receive an explanation of model limitations and a mechanism to record disagreement or data corrections. The employee should not be required to explain a predicted intention that the system has inferred statistically.

Because the supplied application uses broad cross-origin access and lacks demonstrated authentication controls, its current role is a local prototype. A protected deployment would require identity checks, scoped permissions, encrypted connections, audit events, and appropriate data retention. These requirements are motivated by the implemented endpoints and the sensitivity of their outputs. They should be evaluated independently from predictive performance: an accurate classifier cannot compensate for unrestricted access to employee-level records.

X. DISCUSSION

The experiment reveals a practical tension among ranking quality, fixed-threshold detection, and interpretability. Gradient Boosting leads on accuracy and ROC-AUC, Logistic Regression leads on recall and F1-score, and Random Forest provides an operational compromise with efficient TreeSHAP explanations. RQ1 is answered by the different precision–recall tradeoffs. RQ2 is answered by the dominance of overtime, stock-option level, age, income, and manager tenure in global importance. RQ3 is addressed through an interface that keeps prediction, explanation, simulation, and action logging connected under human oversight.

Future work should add repeated cross-validation, calibration, threshold optimization, subgroup fairness tests, temporal drift monitoring, and explanation-stability analysis. Any real deployment also requires privacy controls and a prospective study of whether supportive interventions improve outcomes without harming employees.

A. Synthesis of Evidence

The main finding is that the four candidate models support different operating objectives. A majority-class baseline provides high ordinary accuracy with no positive detection. The linear baseline detects more positives at the cost of a larger review queue. Gradient Boosting produces fewer false alarms but misses more positives. Random Forest lies between those points. The cost sensitivity analysis makes that tradeoff explicit, while the SHAP analysis adds information about the selected model's use of features.

This evidence supports an inspectable prototype and a reproducible comparison of stored results. It does not establish that the selected model is optimal, that its probabilities are calibrated, that its explanations are stable, or that an intervention will reduce turnover. Those unanswered questions are research opportunities rather than reasons to replace measured evidence with optimistic claims. The implementation is useful precisely because its intermediate artifacts allow the claims to be separated and tested.

B. Validation Priorities

The first priority is a consistent evaluation procedure. Repeated stratified validation would characterize sampling variability; an untouched final test set would support reporting after configuration selection. A temporal holdout would become necessary when dated organizational records are available. Threshold selection should use a declared objective and validation data, and the final test results should not be used repeatedly to adjust the decision rule.

The second priority is explanation and simulation consistency. The original and modified profiles should use identical artifacts, transformed names should come from the fitted preprocessor, and the full SHAP decomposition should be checked against the explained output. Plausibility constraints would prevent impossible scenarios. A user study could then assess whether reviewers understand probability, percentage-point change, and attribution without converting them into causal conclusions.

The third priority is outcome evaluation. A prospective study would define an eligible employee cohort, supportive intervention protocol, consent process, observation horizon, and comparison strategy. It would measure employee experience and unintended effects alongside retention outcomes. Until such a study exists, the action tracker provides workflow evidence but cannot demonstrate beneficial organizational impact. This distinction preserves the value of the prototype while identifying what additional evidence a stronger research claim would require.

C. Reporting and Interpretation Protocol

A useful reporting protocol should distinguish training observations, test observations, and demonstration

records. Training observations determine fitted parameters and transformations. Test observations support the reported estimates of predictive behavior. Demonstration records populate the user interface and can include both partitions. A screenshot of a plausible employee profile therefore establishes that the interface can display a result, but does not add an independent observation to the test sample. The same principle applies to explanations computed across the full dataset.

Model comparisons also require a consistent positive class and unit convention. All results here use Attrition=Yes as the positive outcome. The confusion matrices place actual outcomes on rows and predictions on columns. Percentage-valued metrics describe sample proportions, whereas SHAP magnitude describes contributions in the explained model-output space. The simulator's probability difference is expressed in percentage points. Keeping these conventions explicit prevents readers from comparing quantities that have similar-looking numerical values but different meanings.

Finally, operational success should be assessed separately from statistical success. A service can correctly return a prediction while using stale artifacts or an incomplete employee profile. A model can rank test observations reasonably well while its high-risk category remains unsuitable for a particular review capacity. An action can be recorded successfully while its real-world outcome is unknown. These distinctions motivate a validation plan with separate checks for prediction quality, artifact consistency, interface interpretation, and intervention outcomes. They also explain why the present research contribution is an integrated, inspectable decision-support prototype rather than evidence of an achieved reduction in employee turnover.

XI. CONCLUSION

This paper presents an explainable attrition-risk decision-support system that combines reproducible model evaluation, TreeSHAP interpretation, and a full-stack application. The results demonstrate why model selection must consider precision, recall, and operational context rather than accuracy alone. The system makes evidence inspectable while preserving human responsibility and clearly separating model attribution from causal or disciplinary judgment.

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